Job Tracker Backend Documentation

Covers authentication, database layer, REST API, middleware, background jobs, and email delivery.

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1) Executive Summary

Job Tracker is a **Node.js + Express** REST API that powers a full-stack job application management system. The server is a standalone process decoupled from the React frontend, communicating exclusively over HTTP.

At a high level:

- **Express 5** handles routing, middleware, and HTTP request/response lifecycle.
- **MongoDB + Mongoose** persists user accounts and job application records.
- **JWT** tokens issued on login are stored in **HttpOnly cookies**, providing secure, stateless session management.
- **bcrypt** hashes all user passwords before storage; plaintext passwords are never persisted.
- **node-cron** schedules a daily job at 9 AM that scans for due follow-up dates and sends email reminders.
- **Nodemailer** delivers reminder emails through Gmail SMTP.
- **Helmet**, **CORS**, and **express-rate-limit** harden the API against common web threats.

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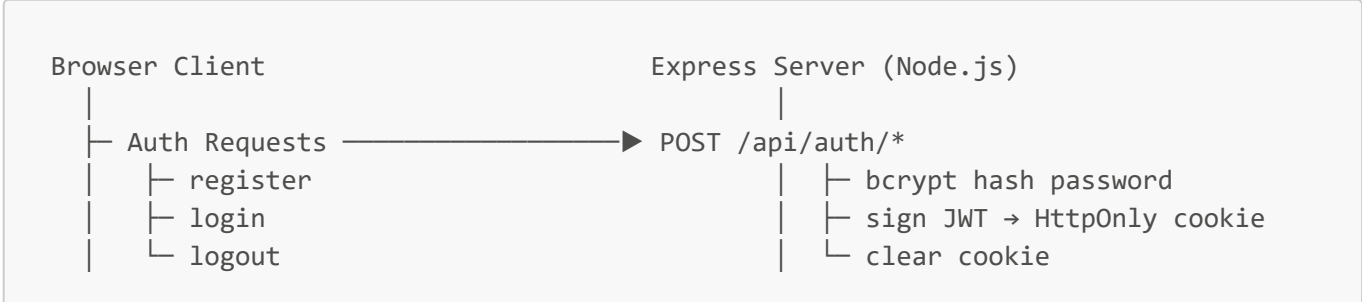
2) Architecture Overview

2.1 System Context



```
  │                                          │
  ├─ Application CRUD ──────────────────▶ /api/applications/*
  │   (JWT cookie sent automatically)        │   ├─ protect middleware validates
token
  │   ├─ GET    /                           │   ├─ ownership check on mutating
ops
  │   ├─ POST   /                           │   └─ Mongoose CRUD
  │   ├─ PUT    /:id                        │
  │   └─ DELETE /:id                         │
  │                                          ▼
  └─ TradingView / Client UI          MongoDB Atlas
                                        ├─ users        (email, hashed password)
                                        └─ applications (all job application
fields)

node-cron (internal scheduler)
  └─ 0 9 * * *  ──────────────────────▶ sendReminders()
                                          ├─ query applications where followUpDate
<= today
                                          │  AND reminderSent = false
                                          ├─ send email via Nodemailer (Gmail
SMTP)
                                          └─ mark reminderSent = true
```

2.2 Runtime Model

The server is a single long-running Node.js process. There is no separate job worker — the cron scheduler (`node-cron`) runs inside the same process and is registered at startup via `server.js`.

Key design decisions:

- **Stateless JWT auth** — No session store is needed. The JWT payload carries `userId`; the `protect` middleware verifies the signature and loads the user from MongoDB on each request.
- **Dual token source** — The `protect` middleware accepts a token from either the `Authorization: Bearer` header or the `token` HttpOnly cookie, supporting both browser clients (cookie) and API clients (header).
- **Resource ownership enforcement** — `updateApplication` and `deleteApplication` perform an explicit ownership check (`application.user === req.user._id`) after fetching the document, preventing users from modifying each other's data.
- **Rate limiting scoped to auth routes** — The limiter is applied only to `/api/auth` to throttle brute-force login and registration attempts without impacting application data endpoints.

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3) Technology Stack

3.1 Core Platform

Package	Version	Role
Node.js	LTS	Runtime

Package	Version	Role
Express	5.x	HTTP framework — routing, middleware, request handling

3.2 Data Layer

Package	Version	Role
MongoDB (Atlas)	—	Primary database
Mongoose	9.x	ODM — schema definition, validation, querying

3.3 Authentication and Security

Package	Version	Role
jsonwebtoken	9.x	JWT signing and verification
bcrypt	6.x	Password hashing (10 salt rounds)
cookie-parser	1.x	Parses `token` cookie from incoming requests
helmet	8.x	Sets secure HTTP response headers
cors	2.x	Cross-origin request policy with allowlist
express-rate-limit	8.x	Rate limiting for auth endpoints

3.4 Background Jobs and Email

Package	Version	Role
node-cron	4.x	In-process cron scheduler
nodemailer	7.x	SMTP email transport via Gmail

3.5 Utility

Package	Role
dotenv	Environment variable loading from `.env`
nodemon	Dev-only — auto-restarts server on file changes

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4) Project Structure

```
server/
  server.js                    # Entry point: loads env, connects DB, starts
cron, binds port
  app.js                       # Express app: global middleware, route mounting
  config/
    db.js                      # Mongoose connection function
```

```
    mailer.js                  # Shared Nodemailer transporter instance
  controllers/
    auth.controller.js         # register, login, logout handlers
    application.controller.js  # createApplication, getApplications,
updateApplication, deleteApplication
  middleware/
    auth.middleware.js         # protect — JWT verification via header or cookie
    validateObjectId.js        # Guards /:id routes against malformed MongoDB
ObjectIds
  models/
    User.js                    # User schema: email, hashed password
    Application.js             # Application schema: all job tracking fields
  routes/
    auth.routes.js             # POST /register, /login, /logout
    application.routes.js      # CRUD routes — all protected
  jobs/
    reminder.job.js            # Cron job: daily follow-up email reminders
```

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5) Data Model

5.1 `users` Collection

Managed by Mongoose via `User.js`. Stores registered user credentials.

Field	Type	Constraints	Notes
`email`	String	Required, unique, indexed	Lowercased and trimmed before save
`password`	String	Required	bcrypt hash — plaintext is never stored
`createdAt`	Date	Auto	Mongoose timestamps
`updatedAt`	Date	Auto	Mongoose timestamps

5.2 `applications` Collection

Managed by Mongoose via `Application.js`. Stores all job application records for all users.

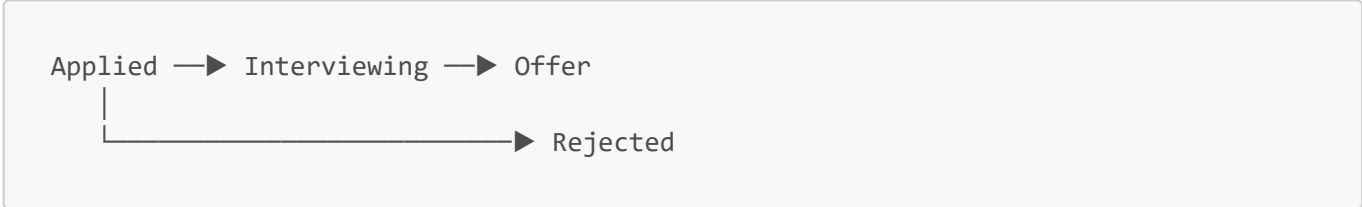
Field	Type	Constraints	Notes
`user`	ObjectId	Required, ref: `User`	Owner of the application
`company`	String	Required, trimmed	Company name
`role`	String	Required, trimmed	Job title / role applied for
`status`	String	Enum, default: `"Applied"`	One of: `Applied`, `Interviewing`, `Offer`, `Rejected`
`appliedDate`	Date	Default: `Date.now`	Date the application was submitted
`followUpDate`	Date	Optional	Date on which a follow-up reminder should be sent

Field	Type	Constraints	Notes
`notes`	String	Optional	Free-text notes about the application
`jobLink`	String	Optional	URL to the job posting
`reminderSent`	Boolean	Default: `false`	Flipped to `true` after the cron job sends an email reminder
`createdAt`	Date	Auto	Mongoose timestamps
`updatedAt`	Date	Auto	Mongoose timestamps

Indexes:

- `user` — implicitly indexed via the ObjectId ref; all queries filter by `user: req.user._id`
- `{ followUpDate, reminderSent }` — queried together by the reminder cron job

5.3 Application Status Lifecycle



Status transitions are not enforced server-side — the client may set any valid enum value freely.

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6) Authentication

File: server/controllers/auth.controller.js

6.1 Token Strategy

Authentication uses **JWT** tokens stored in **HttpOnly cookies**. The token is signed with `JWT_SECRET` and expires after **7 days**. Because the cookie is HttpOnly, it is inaccessible to JavaScript running in the browser, preventing XSS-based token theft.

```
register / login
  └─ generateToken(userId)
       └─ jwt.sign({ userId }, JWT_SECRET, { expiresIn: '7d' })
  └─ setAuthCookie(res, token)
       └─ res.cookie('token', token, {
             httpOnly: true,
             secure: true (production),
             sameSite: 'strict' (production) | 'lax' (development),
             maxAge: 7 days
           })
```

6.2 Auth Settings

Setting	Value	Notes
Token type	JWT	Signed with `HS256` (jsonwebtoken default)
Token lifetime	7 days	Controlled by `expiresIn: '7d'` and `maxAge` cookie option
Password hashing	bcrypt, 10 salt rounds	Applied at registration; compared at login
Cookie name	`token`	HttpOnly, not accessible to client-side JS
`secure` flag	`true` in production	Enforces HTTPS-only cookie transmission
`sameSite`	`strict` in production, `lax` in development	Mitigates CSRF

6.3 Controller Functions

Function	Endpoint	Description
`register`	`POST /api/auth/register`	Validates input, checks for duplicate email, hashes password with bcrypt, creates user, signs JWT, sets auth cookie
`login`	`POST /api/auth/login`	Looks up user by email, compares password hash with bcrypt, signs JWT, sets auth cookie
`logout`	`POST /api/auth/logout`	Clears the `token` cookie with matching options

6.4 Registration Flow

```
POST /api/auth/register
  │
  ├─ Validate: email + password present
  ├─ Check: User.findOne({ email }) → 400 if exists
  ├─ bcrypt.hash(password, 10)
  ├─ User.create({ email, hashedPassword })
  ├─ generateToken(user._id)
  ├─ setAuthCookie(res, token)
  └─ 201 { message: "Registered" }
```

6.5 Login Flow

```
POST /api/auth/login
  │
  ├─ User.findOne({ email }) → 400 if not found
  ├─ bcrypt.compare(password, user.password) → 400 if no match
  ├─ generateToken(user._id)
```

```
    ├─ setAuthCookie(res, token)
    └─ 200 { message: "Logged in" }
```

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7) Middleware

7.1 `protect` — JWT Auth Guard

File: server/middleware/auth.middleware.js

Applied as a router-level middleware to all `/api/applications` routes. Blocks unauthenticated requests before they reach any controller.

Token resolution order:

1. Check `Authorization` header for `Bearer <token>` — used by non-browser API clients.
2. Fall back to `req.cookies.token` — used by the browser client.
3. If neither is present, respond `401 No token provided`.

After verifying the JWT signature, the middleware fetches the user from MongoDB (`User.findById`) and attaches it to `req.user` (excluding the password field). If the token is valid but the user no longer exists, the request is rejected with `401 Not authorized`.

Condition	Response
Valid token, user found	Calls `next()` — `req.user` is populated
Valid token, user not in DB	`401 Not authorized`
Invalid or expired token	`401 Not authorized`
No token at all	`401 No token provided`

7.2 `validateObjectId` — Route Parameter Guard

File: server/middleware/validateObjectId.js

Applied before `updateApplication` and `deleteApplication`. Checks that `req.params.id` is a valid MongoDB ObjectId using `mongoose.Types.ObjectId.isValid()`. Returns `400 Invalid ID` immediately if the check fails, preventing Mongoose from throwing a `CastError` inside the controller.

7.3 `authLimiter` — Rate Limiter

File: server/app.js

Scoped to all `/api/auth` routes. Limits each IP to **20 requests per 15-minute window** using `express-rate-limit`. Exceeding the limit returns a `429 Too Many Requests` response with standard `RateLimit-*` headers.

Setting	Value
Window	15 minutes

Setting	Value
Max requests	20 per IP
Headers	Standard (`RateLimit-*`), legacy headers disabled

7.4 Global Middleware (app.js)

Middleware	Purpose
`helmet()`	Sets 15+ secure HTTP headers (CSP, HSTS, X-Frame-Options, etc.)
`cors({ origin, credentials })`	Restricts origins to the `CORS_ORIGIN` allowlist; allows cookies via `credentials: true`
`cookieParser()`	Populates `req.cookies` so the auth middleware can read the `token` cookie
`express.json()`	Parses JSON request bodies into `req.body`

8) API Reference

Base URL: `/api`

All `/api/applications` routes require a valid `token` cookie or `Authorization: Bearer <token>` header.

8.1 Auth Routes

Rate-limited: 20 requests per 15 minutes per IP.

`POST /api/auth/register`

Creates a new user account and sets an auth cookie.

Request body:

```
{
  "email": "user@example.com",
  "password": "securepassword"
}
```

Responses:

Status	Body	Condition
`201`	`{ "message": "Registered" }`	User created successfully
`400`	`{ "message": "All fields are required" }`	Missing email or password
`400`	`{ "message": "User already exists" }`	Email already in use

Status	Body	Condition
500	{ "message": "Server error" }	Unexpected server failure

`POST /api/auth/login`

Authenticates an existing user and sets an auth cookie.

Request body:

```
{
  "email": "user@example.com",
  "password": "securepassword"
}
```

Responses:

Status	Body	Condition
200	{ "message": "Logged in" }	Auth cookie set
400	{ "message": "Invalid credentials" }	Email not found or password mismatch
500	{ "message": "Server error" }	Unexpected server failure

`POST /api/auth/logout`

Clears the auth cookie.

Responses:

Status	Body
200	{ "message": "Logged out" }

8.2 Application Routes

All routes require authentication. The `protect` middleware runs on the entire router.

`GET /api/applications`

Returns all job applications belonging to the authenticated user, sorted by creation date descending.

Responses:

Status	Body
200	Array of application objects

Status	Body
401	{ "message": "Not authorized" }
500	{ "message": "Server error" }

Example response item:

```
{
  "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
  "user": "64f1a2b3c4d5e6f7a8b9c0d0",
  "company": "Acme Corp",
  "role": "Software Engineer",
  "status": "Interviewing",
  "appliedDate": "2025-03-01T00:00:00.000Z",
  "followUpDate": "2025-03-10T00:00:00.000Z",
  "notes": "Spoke with recruiter on 3/5",
  "jobLink": "https://acme.com/jobs/123",
  "reminderSent": false,
  "createdAt": "2025-03-01T12:00:00.000Z",
  "updatedAt": "2025-03-05T09:00:00.000Z"
}
```

`POST /api/applications`

Creates a new job application for the authenticated user.

Request body (all optional fields may be omitted):

```
{
  "company": "Acme Corp",
  "role": "Software Engineer",
  "status": "Applied",
  "appliedDate": "2025-03-01",
  "followUpDate": "2025-03-10",
  "notes": "Applied through LinkedIn",
  "jobLink": "https://acme.com/jobs/123"
}
```

Field	Required	Notes
company	Yes	
role	Yes	
status	No	Defaults to "Applied"
appliedDate	No	Defaults to current date

Field	Required	Notes
`followUpDate`	No	When set, the cron job will send a reminder on this date
`notes`	No	
`jobLink`	No	

Responses:

Status	Body	Condition
`201`	Created application object	Success
`401`	`{ "message": "Not authorized" }`	Missing or invalid token
`500`	`{ "message": "Server error" }`	Unexpected server failure

`PUT /api/applications/:id`

Updates an existing application. Only the owning user may update their own applications.

Middleware chain: `protect` → `validateObjectId` → `updateApplication`

Request body: Any subset of application fields to update.

Responses:

Status	Body	Condition
`200`	Updated application object	Success
`400`	`{ "message": "Invalid ID" }`	`id` is not a valid ObjectId
`401`	`{ "message": "Not authorized" }`	Token invalid or user does not own this application
`404`	`{ "message": "Application not found" }`	No document with that `id`
`500`	`{ "message": "Server error" }`	Unexpected server failure

`DELETE /api/applications/:id`

Permanently deletes an application. Only the owning user may delete their own applications.

Middleware chain: `protect` → `validateObjectId` → `deleteApplication`

Responses:

Status	Body	Condition
`200`	`{ "message": "Application removed" }`	Deleted successfully

Status	Body	Condition
400	{ "message": "Invalid ID" }	id is not a valid ObjectId
401	{ "message": "Not authorized" }	Token invalid or user does not own this application
404	{ "message": "Application not found" }	No document with that id
500	{ "message": "Server error" }	Unexpected server failure

9) Background Jobs

File: server/jobs/reminder.job.js

The reminder job is the only background task in the system. It is registered at server startup by server.js via require('./jobs/reminder.job') and runs inside the main Node.js process using node-cron.

9.1 sendReminders

Schedule: 0 9 * * * — every day at 9:00 AM server local time.

Logic:

Step	Description
1	Construct today as a Date with time set to 00:00:00.000
2	Query Application.find({ followUpDate: { $lte: today }, reminderSent: false }) with .populate('user') to get the user's email
3	For each matching application: send a reminder email via Nodemailer, then set reminderSent = true and save the document
4	Log the count of emails sent

Email format:

Field	Value
From	process.env.EMAIL_USER
To	application.user.email
Subject	Follow-up Reminder: <company>
Body	Plain text: Reminder to follow up on your <role> application at <company>.

Idempotency: The reminderSent flag ensures each application triggers at most one reminder email, even if the job runs multiple times in a day or if the server restarts.

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10) Email System

File: server/config/mailer.js

10.1 Transport

A single shared Nodemailer transporter is created at module load time and exported for use by the reminder job.

```
nodemailer.createTransport({
    service: 'gmail',
    auth: { user: process.env.EMAIL_USER, pass: process.env.EMAIL_PASS }
})
```

The transporter uses Gmail's built-in SMTP settings. `EMAIL_PASS` must be a **Gmail App Password**, not the account's login password.

10.2 Email Types

Trigger	Subject	Content
Follow-up date reached (cron)	`Follow-up Reminder: <company>`	Plain text reminder to follow up on the specific application

11) Configuration and Environment

Required Variables

Variable	Description
`MONGO_URI`	MongoDB Atlas connection string
`JWT_SECRET`	Secret key used to sign and verify JWT tokens (use a long, random string)
`EMAIL_USER`	Gmail address used as the SMTP sender account
`EMAIL_PASS`	Gmail App Password for the sender account (not the account login password)
`CORS_ORIGIN`	Comma-separated list of allowed frontend origins (e.g. `http://localhost:5173`)
`PORT`	Port the Express server listens on (defaults to `5000` if not set)
`NODE_ENV`	Set to `production` to enable secure cookies and strict `sameSite` policy

Development Server Commands

```
# Install dependencies
npm install

# Start server with hot-reload (nodemon)
npm run dev
```

```
# Start server (production)
npm start
```

Security Notes for Production

- Set `NODE_ENV=production` to enable `secure: true` and `sameSite: 'strict'` on the auth cookie.
- If the frontend and backend are deployed on **different domains**, update the cookie options to `sameSite: 'none'` and `secure: true`, and ensure `CORS_ORIGIN` includes the exact frontend domain.
- Use a strong, randomly generated `JWT_SECRET` (minimum 32 characters). Never commit it to source control.
- Gmail App Passwords require 2-Factor Authentication to be enabled on the sender Google account.

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12) Component-by-Component Reference

`server/server.js`

Entry point. Calls `dotenv.config()` first to ensure environment variables are available before any other module loads. Connects to MongoDB, registers the cron job by requiring `reminder.job.js`, then starts the HTTP server on `PORT`.

`server/app.js`

Configures and exports the Express application. Applies global middleware (Helmet, CORS, cookie-parser, JSON body parser), attaches the rate limiter to the auth router, and mounts the two route modules at `/api/auth` and `/api/applications`.

`server/config/db.js`

Exports `connectDB`, an async function that calls `mongoose.connect()` with the `MONGO_URI` environment variable. Logs success or exits the process with code `1` on connection failure, preventing the server from running without a database.

`server/config/mailer.js`

Creates and exports a single shared Nodemailer transporter configured for Gmail SMTP. Instantiated once at module load; imported by the reminder job.

`server/models/User.js`

Defines the `User` Mongoose schema. Email is enforced as unique and lowercased. Password is stored as a bcrypt hash — the schema itself applies no transformation; hashing is done in the controller before `User.create()`.

`server/models/Application.js`

Defines the `Application` Mongoose schema. The `user` field holds the ObjectId of the owning user, enforcing data isolation. The `status` field is constrained to four enum values. The `reminderSent` flag controls the cron job's idempotency. Includes `timestamps: true` for automatic `createdAt` / `updatedAt` fields.

server/middleware/auth.middleware.js

Exports the `protect` middleware. Checks the `Authorization` header first, then the `token` cookie. Verifies the JWT with `jwt.verify()`, loads the user from MongoDB, and attaches the user (without password) to `req.user`. Rejects with `401` on any failure.

server/middleware/validateObjectId.js

Exports `validateObjectId`. A lightweight guard that calls `mongoose.Types.ObjectId.isValid(req.params.id)` and returns `400 Invalid ID` on failure, keeping controller code free of ObjectId casting concerns.

server/controllers/auth.controller.js

Implements `register`, `login`, and `logout`. Defines the `generateToken` helper (signs a JWT with `userId` payload) and the `setAuthCookie` helper (sets cookie with environment-aware security flags). Both helpers are private to this module.

server/controllers/application.controller.js

Implements the four CRUD handlers. All handlers scope queries to `req.user._id` provided by the auth middleware. `updateApplication` and `deleteApplication` perform an explicit ownership check after fetching the document, returning `401` if the document's `user` field does not match the requester's ID.

server/routes/auth.routes.js

Mounts `register`, `login`, and `logout` on three `POST` routes. No auth middleware — these endpoints must be publicly accessible.

server/routes/application.routes.js

Applies `protect` at the router level via `router.use(protect)`, so every route in this file is automatically authenticated. `PUT` and `DELETE` also chain `validateObjectId` before the controller.

server/jobs/reminder.job.js

Exports `sendReminders` and immediately schedules it with `cron.schedule('0 9 * * *', sendReminders)`. The job queries for all applications with an overdue, unsent follow-up date, sends each user a plain-text reminder email, and marks each application as `reminderSent: true` to prevent re-sending.